

Отчёт по лабораторной работе 8

Адресация IPv4 и IPv6. Настройка маршрутизации

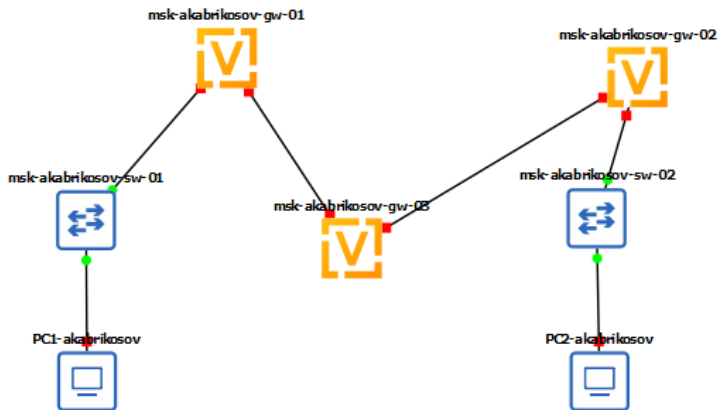
Абрикосов Артём

Российский университет дружбы народов, Москва, Россия

Цели и задачи работы

Изучение принципов маршрутизации в IPv4- и IPv6-сетях и принципов настройки сетевого оборудования в среде GNS3.

Выполнение лабораторной работы



Настройка IPv6 на PC1

```
PC1-akabrikosov - PuTTY
Executing the startup file

Hostname is too long. (Maximum 12 characters)

VPCS> ip 1000::a/64
PC1 : 1000::a/64

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ipv6

NAME                : VPCS[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6800/64
GLOBAL SCOPE        : 1000::a/64
DNS                  :
ROUTER LINK-LAYER   :
MAC                  : 00:50:79:66:68:00
LPORT                : 10012
RHOST:PORT           : 127.0.0.1:10013
MTU                  : 1500

VPCS>
```

Настройка IPv6 на PC2

```
PC2-akabrikosov - PuTTY
Executing the startup file

Hostname is too long. (Maximum 12 characters)

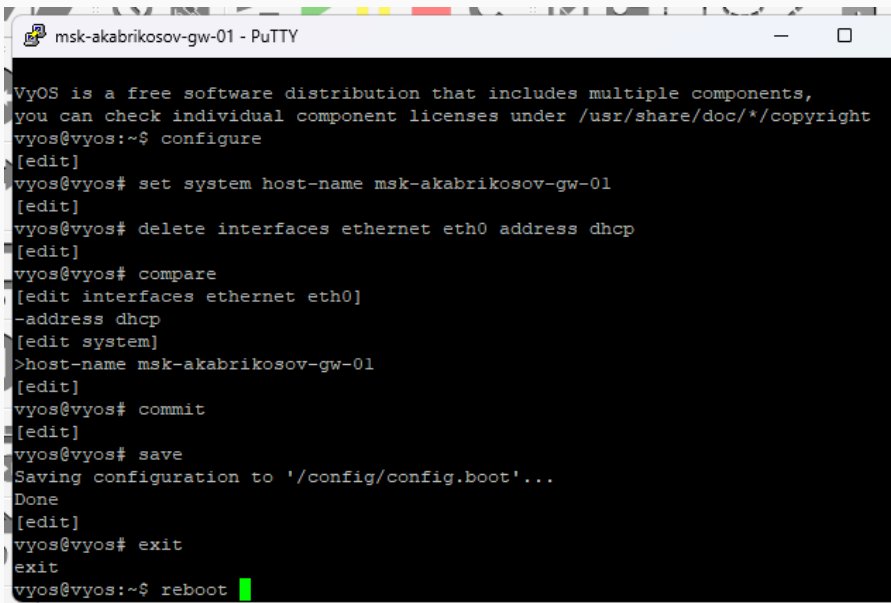
VPCS> ip 1002::a/64
PC1 : 1002::a/64

VPCS> save
Saving startup configuration to startup.vpc
. done

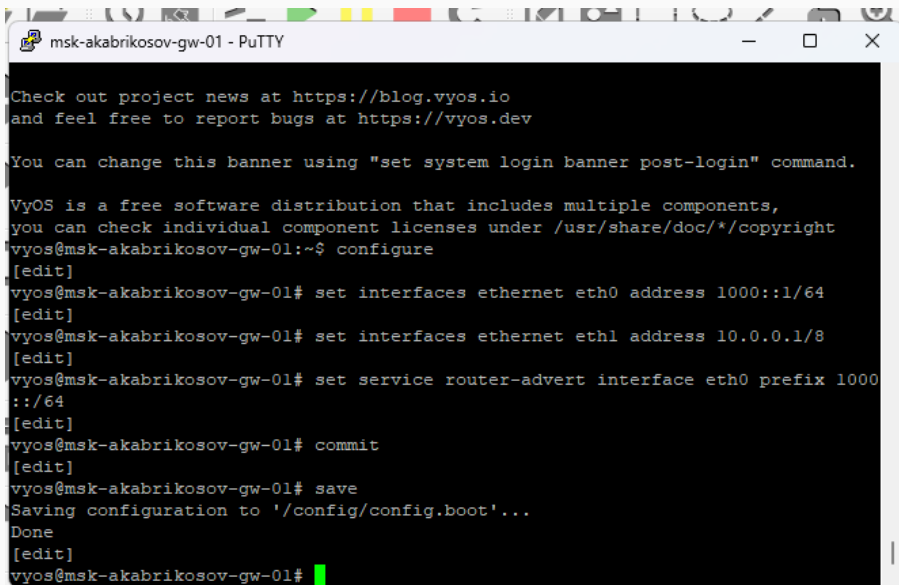
VPCS> show ipv6

NAME                : VPCS[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6801/64
GLOBAL SCOPE        : 1002::a/64
DNS                 :
ROUTER LINK-LAYER   :
MAC                 : 00:50:79:66:68:01
LPORT               : 10014
RHOST:PORT           : 127.0.0.1:10015
MTU                 : 1500

VPCS>
```



```
msk-akabrikosov-gw-01 - PuTTY
VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
vyos@vyos:~$ configure
[edit]
vyos@vyos# set system host-name msk-akabrikosov-gw-01
[edit]
vyos@vyos# delete interfaces ethernet eth0 address dhcp
[edit]
vyos@vyos# compare
[edit interfaces ethernet eth0]
-address dhcp
[edit system]
>host-name msk-akabrikosov-gw-01
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# exit
exit
vyos@vyos:~$ reboot
```



```
msk-akabrikosov-gw-01 - PuTTY

Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

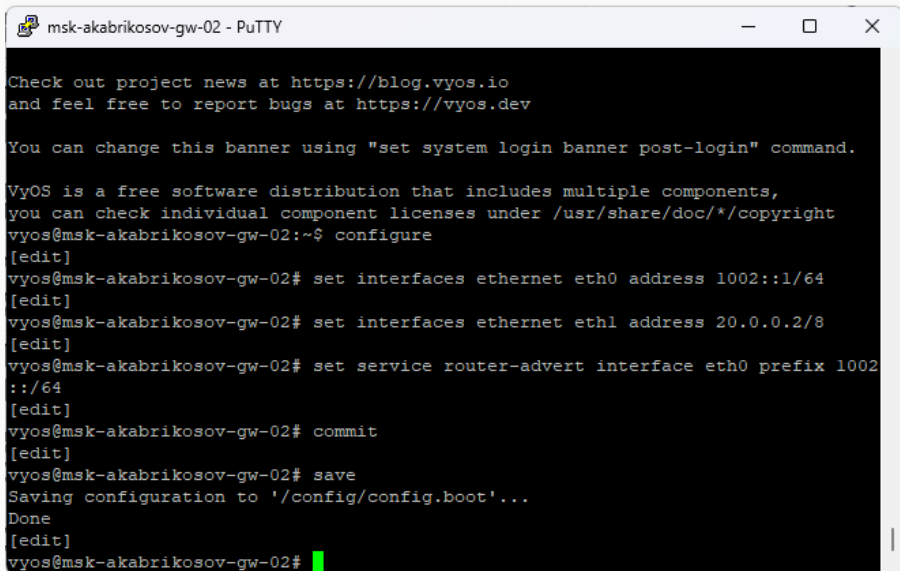
VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
vyos@msk-akabrikosov-gw-01:~$ configure
[edit]
vyos@msk-akabrikosov-gw-01# set interfaces ethernet eth0 address 1000::1/64
[edit]
vyos@msk-akabrikosov-gw-01# set interfaces ethernet eth1 address 10.0.0.1/8
[edit]
vyos@msk-akabrikosov-gw-01# set service router-advert interface eth0 prefix 1000
::/64
[edit]
vyos@msk-akabrikosov-gw-01# commit
[edit]
vyos@msk-akabrikosov-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-akabrikosov-gw-01#
```

```
msk-akabrikosov-gw-02 - PuTTY
vyos login: vyos
Password:
Welcome to VyOS!

Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
vyos@vyos:~$ configure
[edit]
vyos@vyos# set system host-name msk-akabrikosov-gw-02
[edit]
vyos@vyos# delete interfaces ethernet eth0 address dhcp
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos#
```



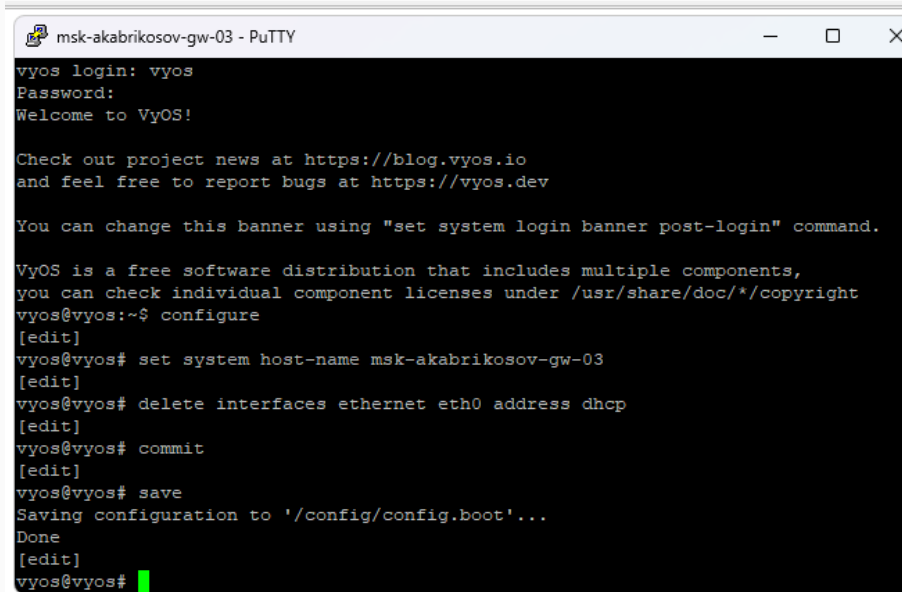
```
msk-akabrikosov-gw-02 - PuTTY

Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
vyos@msk-akabrikosov-gw-02:~$ configure
[edit]
vyos@msk-akabrikosov-gw-02# set interfaces ethernet eth0 address 1002::1/64
[edit]
vyos@msk-akabrikosov-gw-02# set interfaces ethernet eth1 address 20.0.0.2/8
[edit]
vyos@msk-akabrikosov-gw-02# set service router-advert interface eth0 prefix 1002
::/64
[edit]
vyos@msk-akabrikosov-gw-02# commit
[edit]
vyos@msk-akabrikosov-gw-02# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-akabrikosov-gw-02#
```

Подготовка и адресация gw-03



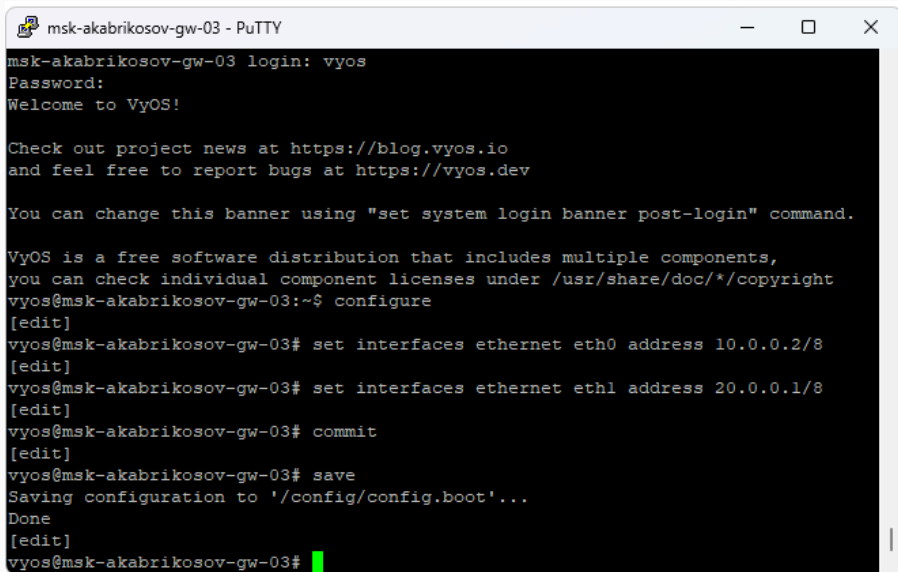
```
msk-akabrikosov-gw-03 - PuTTY
vyos login: vyos
Password:
Welcome to VyOS!

Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
vyos@vyos:~$ configure
[edit]
vyos@vyos# set system host-name msk-akabrikosov-gw-03
[edit]
vyos@vyos# delete interfaces ethernet eth0 address dhcp
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos#
```

Подготовка и адресация gw-03

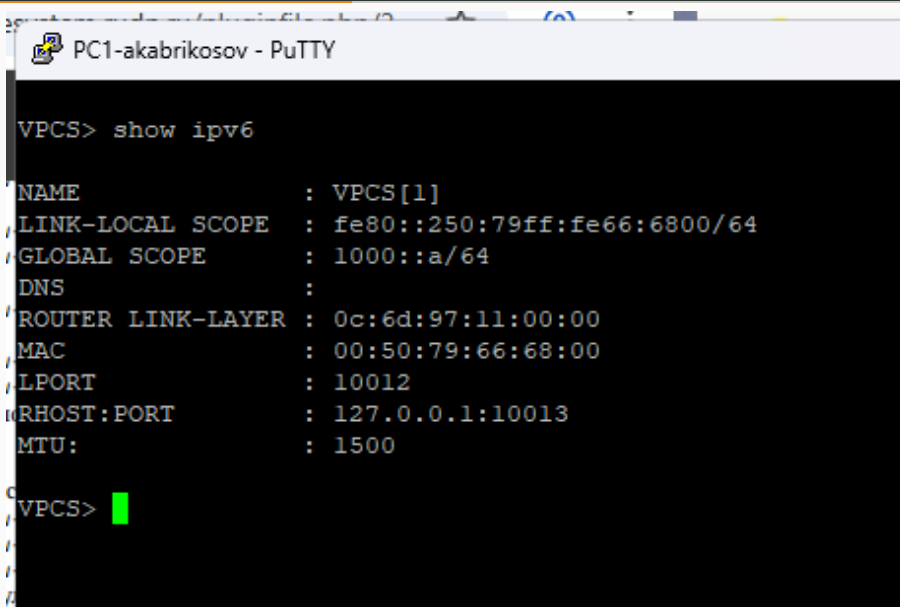


```
msk-akabrikosov-gw-03 login: vyos
Password:
Welcome to VyOS!

Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
vyos@msk-akabrikosov-gw-03:~$ configure
[edit]
vyos@msk-akabrikosov-gw-03# set interfaces ethernet eth0 address 10.0.0.2/8
[edit]
vyos@msk-akabrikosov-gw-03# set interfaces ethernet eth1 address 20.0.0.1/8
[edit]
vyos@msk-akabrikosov-gw-03# commit
[edit]
vyos@msk-akabrikosov-gw-03# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-akabrikosov-gw-03#
```

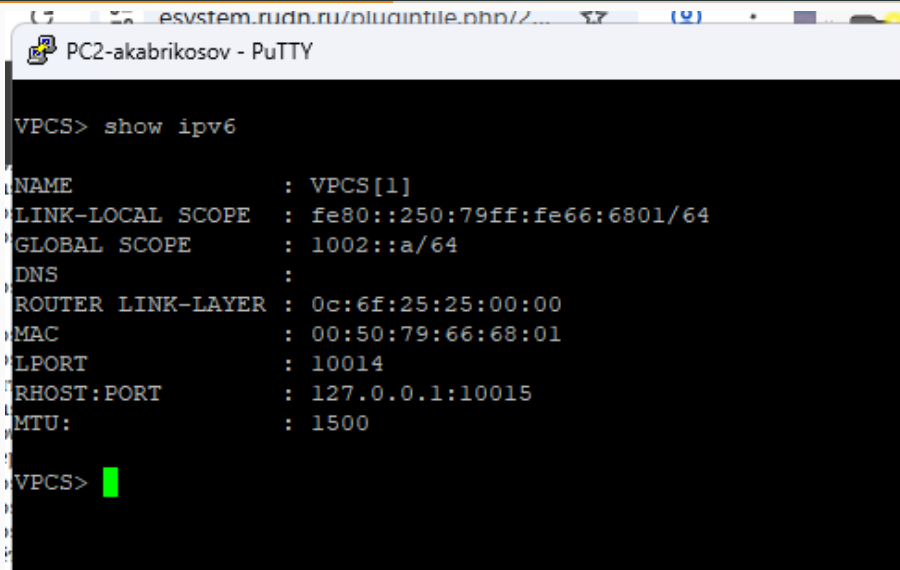


```
PC1-akabrikosov - PuTTY

VPCS> show ipv6

NAME                : VPCS[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6800/64
GLOBAL SCOPE        : 1000::a/64
DNS                 :
ROUTER LINK-LAYER   : 0c:6d:97:11:00:00
MAC                 : 00:50:79:66:68:00
LPORT               : 10012
RHOST:PORT           : 127.0.0.1:10013
MTU                 : 1500

VPCS> 
```



The screenshot shows a PuTTY terminal window titled "PC2-akabrikosov - PuTTY". The terminal displays the output of the command "show ipv6" in a VPCS environment. The output lists various IPv6-related parameters and their values. A red cursor is visible at the end of the last line "VPCS>".

```
VPCS> show ipv6

NAME                : VPCS[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6801/64
GLOBAL SCOPE        : 1002::a/64
DNS                 :
ROUTER LINK-LAYER   : 0c:6f:25:25:00:00
MAC                 : 00:50:79:66:68:01
LPORT               : 10014
RHOST:PORT          : 127.0.0.1:10015
MTU                 : 1500

VPCS>
```

Ping с gw-01 до настройки маршрутизации

```
vyos@msk-akabrikov-gw-01#  
[edit]  
vyos@msk-akabrikov-gw-01# ping 10.0.0.2  
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.  
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=2.53 ms  
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=1.06 ms  
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=3.16 ms  
^C  
--- 10.0.0.2 ping statistics ---  
3 packets transmitted, 3 received, 0% packet loss, time 16ms  
rtt min/avg/max/mdev = 1.064/2.251/3.164/0.880 ms  
[edit]  
vyos@msk-akabrikov-gw-01# ping 20.0.0.1  
connect: Network is unreachable  
[edit]  
vyos@msk-akabrikov-gw-01# ping 20.0.0.2  
connect: Network is unreachable  
[edit]  
vyos@msk-akabrikov-gw-01#
```

Рис. 12: Проверка связности с gw-01

Анализ трафика до RIP (gw-01 ↔ gw-03)

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:6d:97:11:00:01	Broadcast	ARP	60	Who has 10.0.0.2? Tell 10.0.0.1
2	0.001046	0c:24:7d:5a:00:00	0c:6d:97:11:00:01	ARP	60	10.0.0.2 is at 0c:24:7d:5a:00:00
3	0.001402	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request id=0x0829, seq=1/256, ttl=64
4	0.001846	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply id=0x0829, seq=1/256, ttl=64
5	1.001557	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request id=0x0829, seq=2/512, ttl=64
6	1.002293	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply id=0x0829, seq=2/512, ttl=64
7	2.013478	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request id=0x0829, seq=3/768, ttl=64
8	2.015116	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply id=0x0829, seq=3/768, ttl=64
9	5.285957	0c:24:7d:5a:00:00	0c:6d:97:11:00:01	ARP	60	Who has 10.0.0.1? Tell 10.0.0.2
10	5.287645	0c:6d:97:11:00:01	0c:24:7d:5a:00:00	ARP	60	10.0.0.1 is at 0c:6d:97:11:00:01

> Frame 8: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0

> Ethernet II, Src: 0c:24:7d:5a:00:00 (0c:24:7d:5a:00:00), Dst: 0c:6d:97:11:00:01 (0c:6d:97:11:00:01)

▼ Internet Protocol Version 4, Src: 10.0.0.2, Dst: 10.0.0.1

- 0100 = Version: 4
- 0101 = Header Length: 20 bytes (5)
- > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
- Total Length: 84
- Identification: 0x9e85 (40581)
- > 000. = Flags: 0x0
- ...0 0000 0000 0000 = Fragment Offset: 0
- Time to Live: 64
- Protocol: ICMP (1)
- Header Checksum: 0xc821 [validation disabled]

0000 0c 6d 97 11 00 01

0010 00 54 05 00 00 00 00

0020 00 00 00 00 00 00 00

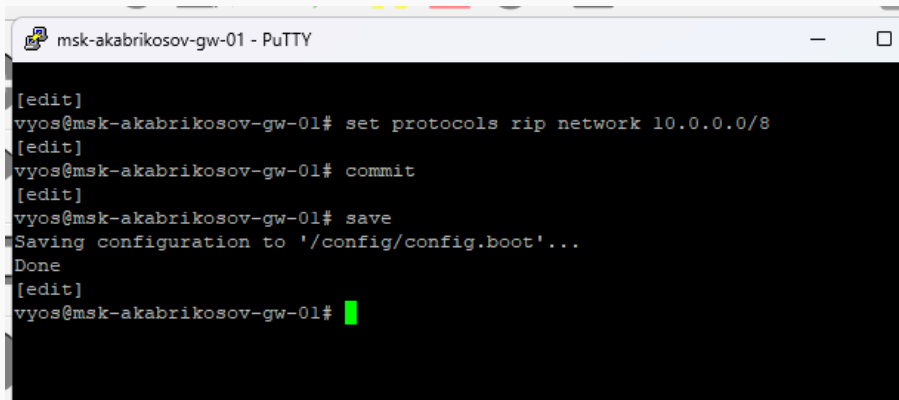
0030 00 00 00 00 00 00 00

0040 16 17 00 00 00 00 00

0050 26 27 00 00 00 00 00

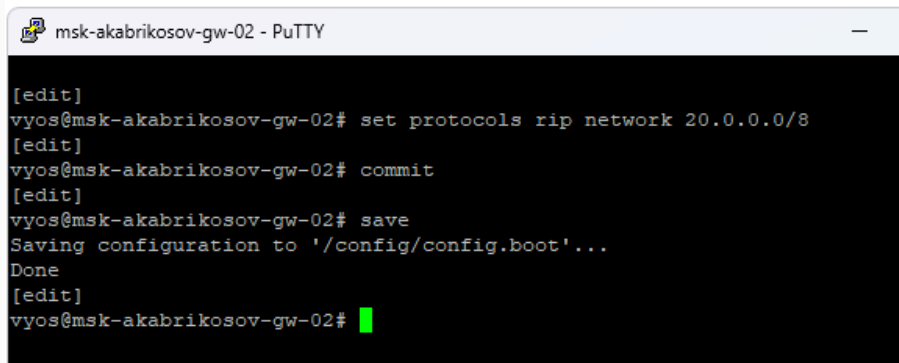
0060 36 37 00 00 00 00 00

Настройка RIP на gw-01



```
msk-akabrikosov-gw-01 - PuTTY
[edit]
vyos@msk-akabrikosov-gw-01# set protocols rip network 10.0.0.0/8
[edit]
vyos@msk-akabrikosov-gw-01# commit
[edit]
vyos@msk-akabrikosov-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-akabrikosov-gw-01#
```

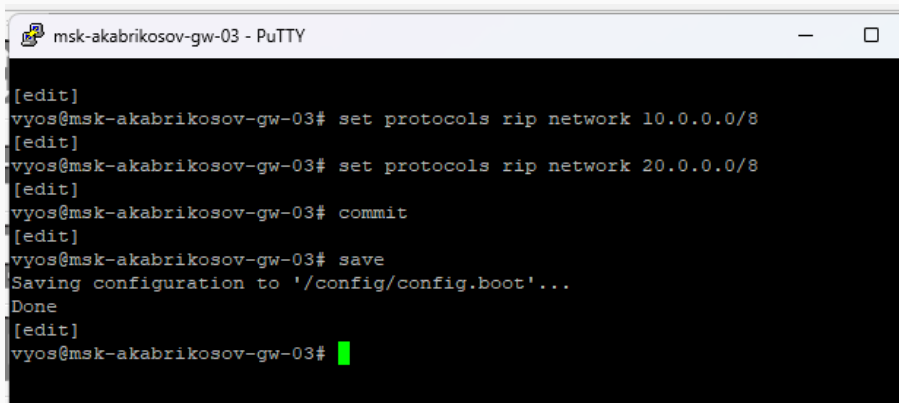
Рис. 14: RIP на gw-01



The screenshot shows a PuTTY terminal window titled "msk-akabrikosov-gw-02 - PuTTY". The terminal displays the following commands and output:

```
[edit]
vyos@msk-akabrikosov-gw-02# set protocols rip network 20.0.0.0/8
[edit]
vyos@msk-akabrikosov-gw-02# commit
[edit]
vyos@msk-akabrikosov-gw-02# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-akabrikosov-gw-02#
```

Рис. 15: RIP на gw-02



```
msk-akabrikosov-gw-03 - PuTTY

[edit]
vyos@msk-akabrikosov-gw-03# set protocols rip network 10.0.0.0/8
[edit]
vyos@msk-akabrikosov-gw-03# set protocols rip network 20.0.0.0/8
[edit]
vyos@msk-akabrikosov-gw-03# commit
[edit]
vyos@msk-akabrikosov-gw-03# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-akabrikosov-gw-03#
```

Рис. 16: RIP на gw-03

Проверка связности после включения RIP

```
msh-akabrikosov-gw-01 - PuTTY

[edit]
vyos@msh-akabrikosov-gw-01# ping 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=1.45 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=4.93 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=2.63 ms
^C
--- 10.0.0.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 11ms
rtt min/avg/max/mdev = 1.445/3.000/4.930/1.447 ms
[edit]
vyos@msh-akabrikosov-gw-01# ping 20.0.0.1
PING 20.0.0.1 (20.0.0.1) 56(84) bytes of data.
64 bytes from 20.0.0.1: icmp_seq=1 ttl=64 time=2.01 ms
64 bytes from 20.0.0.1: icmp_seq=2 ttl=64 time=4.45 ms
64 bytes from 20.0.0.1: icmp_seq=3 ttl=64 time=4.08 ms
^C
--- 20.0.0.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 7ms
rtt min/avg/max/mdev = 2.007/3.510/4.446/1.075 ms
[edit]
vyos@msh-akabrikosov-gw-01# ping 20.0.0.2
PING 20.0.0.2 (20.0.0.2) 56(84) bytes of data.
64 bytes from 20.0.0.2: icmp_seq=1 ttl=63 time=6.42 ms
64 bytes from 20.0.0.2: icmp_seq=2 ttl=63 time=4.88 ms
^C
```

Анализ трафика после RIP

No.	Time	Source	Destination	Protocol	Length	Info
9	19.079914	0c:24:7d:5a:00:00	0c:6d:97:11:00:01	ARP	60	10.0.0.2 is at 0c:24:7d:5a:00:00
10	19.349732	0c:24:7d:5a:00:00	0c:6d:97:11:00:01	ARP	60	Who has 10.0.0.1? Tell 10.0.0.2
11	19.350660	0c:6d:97:11:00:01	0c:24:7d:5a:00:00	ARP	60	10.0.0.1 is at 0c:6d:97:11:00:01
12	23.819838	10.0.0.1	20.0.0.1	ICMP	98	Echo (ping) request id=0x0880, seq=1/256, t
13	23.820965	20.0.0.1	10.0.0.1	ICMP	98	Echo (ping) reply id=0x0880, seq=1/256, t
14	24.822753	10.0.0.1	20.0.0.1	ICMP	98	Echo (ping) request id=0x0880, seq=2/512, t
15	24.824808	20.0.0.1	10.0.0.1	ICMP	98	Echo (ping) reply id=0x0880, seq=2/512, t
16	25.824753	10.0.0.1	20.0.0.1	ICMP	98	Echo (ping) request id=0x0880, seq=3/768, t
17	25.827722	20.0.0.1	10.0.0.1	ICMP	98	Echo (ping) reply id=0x0880, seq=3/768, t
→ 18	30.630484	10.0.0.1	20.0.0.2	ICMP	98	Echo (ping) request id=0x0881, seq=1/256, t
← 19	30.635959	20.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply id=0x0881, seq=1/256, t
20	31.647909	10.0.0.1	20.0.0.2	ICMP	98	Echo (ping) request id=0x0881, seq=2/512, t
21	31.651941	20.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply id=0x0881, seq=2/512, t
22	34.301077	10.0.0.2	224.0.0.9	RIPv2	66	Response
23	60.313191	10.0.0.2	224.0.0.9	RIPv2	66	Response

> Frame 19: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0

> Ethernet II, Src: 0c:24:7d:5a:00:00 (0c:24:7d:5a:00:00), Dst: 0c:6d:97:11:00:01 (0c:6d:97:11:00:01)

> Internet Protocol Version 4, Src: 20.0.0.2, Dst: 10.0.0.1

▼ Internet Control Message Protocol

Type: Echo (ping) reply (0)

Code: 0

Checksum: 0xe959 [correct]

[Checksum Status: Good]

Identifier (BE): 2177 (0x0881)

Identifier (LE): 33032 (0x8108)

Sequence Number (BE): 1 (0x0001)

Sequence Number (LE): 256 (0x0100)

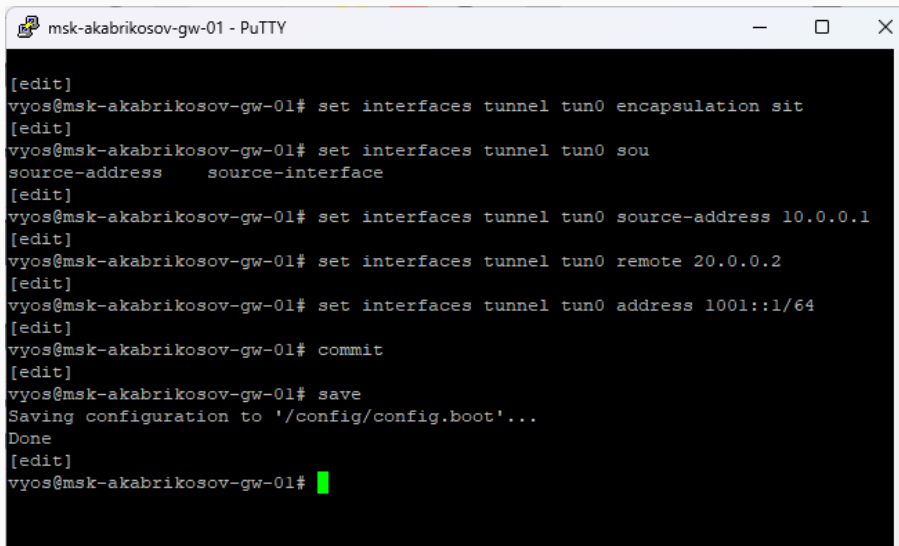
[\[Request frame: 18\]](#)

[Response time: 5,475 ms]

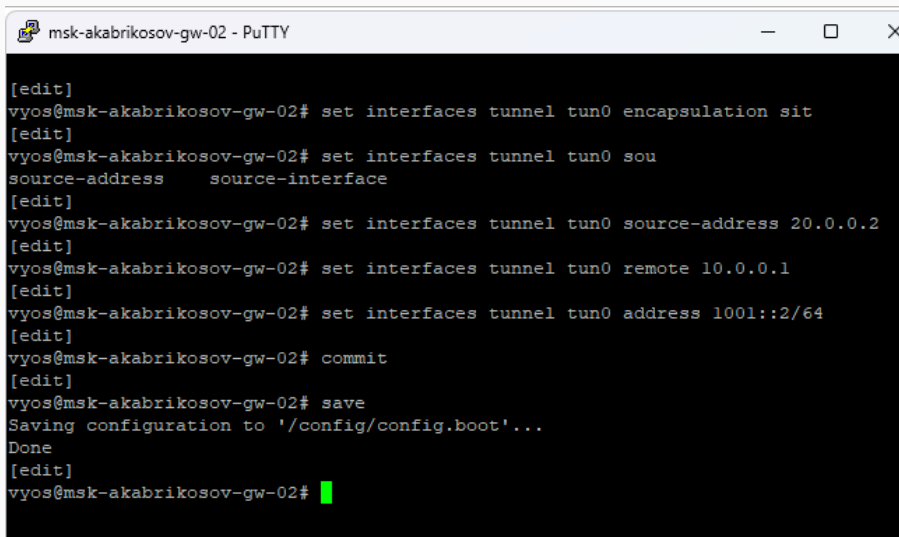
> ICMP Data: 03f7856900000000bbf00a000000000101112131415161718191a1b1c1d1e1f202122232425262728292a2b2c2d2e...

0000	0c 6d
0010	00 54
0020	00 01
0030	00 00
0040	16 17
0050	26 27
0060	36 37

Туннель SIT на gw-01

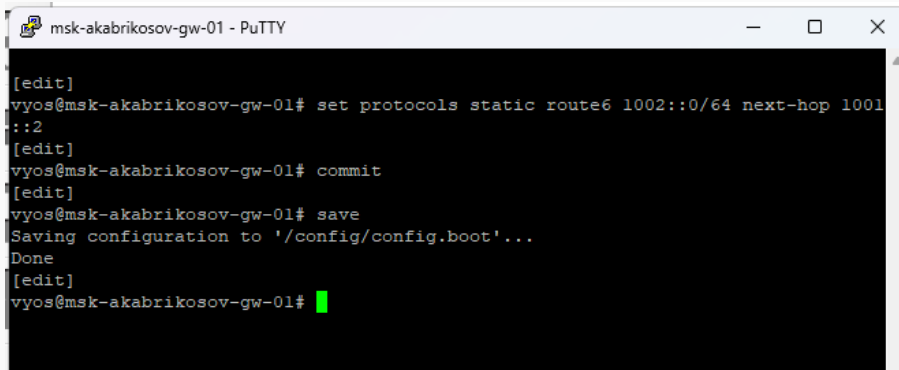


```
[edit]
vyos@msk-akabrikosov-gw-01# set interfaces tunnel tun0 encapsulation sit
[edit]
vyos@msk-akabrikosov-gw-01# set interfaces tunnel tun0 sou
source-address      source-interface
[edit]
vyos@msk-akabrikosov-gw-01# set interfaces tunnel tun0 source-address 10.0.0.1
[edit]
vyos@msk-akabrikosov-gw-01# set interfaces tunnel tun0 remote 20.0.0.2
[edit]
vyos@msk-akabrikosov-gw-01# set interfaces tunnel tun0 address 1001::1/64
[edit]
vyos@msk-akabrikosov-gw-01# commit
[edit]
vyos@msk-akabrikosov-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-akabrikosov-gw-01#
```



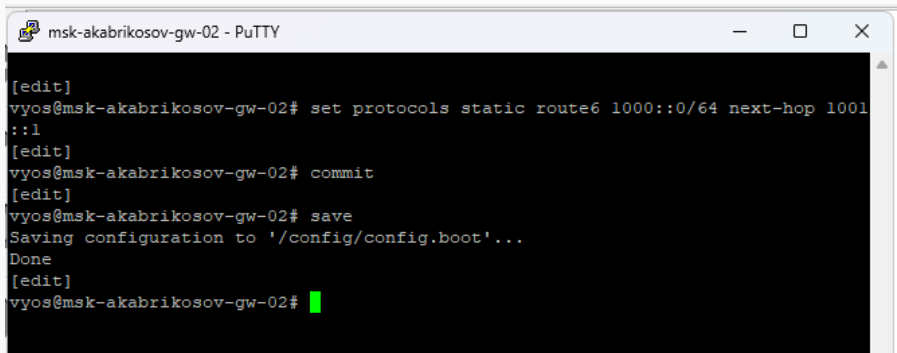
```
[edit]
vyos@msk-akabrikosov-gw-02# set interfaces tunnel tun0 encapsulation sit
[edit]
vyos@msk-akabrikosov-gw-02# set interfaces tunnel tun0 sou
source-address      source-interface
[edit]
vyos@msk-akabrikosov-gw-02# set interfaces tunnel tun0 source-address 20.0.0.2
[edit]
vyos@msk-akabrikosov-gw-02# set interfaces tunnel tun0 remote 10.0.0.1
[edit]
vyos@msk-akabrikosov-gw-02# set interfaces tunnel tun0 address 1001::2/64
[edit]
vyos@msk-akabrikosov-gw-02# commit
[edit]
vyos@msk-akabrikosov-gw-02# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-akabrikosov-gw-02#
```

Рис. 20: Туннель SIT на gw-02



```
msk-akabrikosov-gw-01 - PuTTY
[edit]
vyos@msk-akabrikosov-gw-01# set protocols static route6 1002::0/64 next-hop 1001
:::2
[edit]
vyos@msk-akabrikosov-gw-01# commit
[edit]
vyos@msk-akabrikosov-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-akabrikosov-gw-01#
```

Рис. 21: Статический IPv6-маршрут на gw-01



```
msk-akabrikosov-gw-02 - PuTTY
[edit]
vyos@msk-akabrikosov-gw-02# set protocols static route6 1000::0/64 next-hop 1001
::1
[edit]
vyos@msk-akabrikosov-gw-02# commit
[edit]
vyos@msk-akabrikosov-gw-02# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-akabrikosov-gw-02#
```

Рис. 22: Статический IPv6-маршрут на gw-02

Проверка с PC1: ping и trace до PC2

```
PC1-akabrikosov - PuTTY

VPCS> ping 1002::a

1002::a icmp6_seq=1 ttl=60 time=20.798 ms
1002::a icmp6_seq=2 ttl=60 time=3.388 ms
1002::a icmp6_seq=3 ttl=60 time=10.278 ms
1002::a icmp6_seq=4 ttl=60 time=9.678 ms
1002::a icmp6_seq=5 ttl=60 time=11.138 ms

VPCS> trace 1002::a

trace to 1002::a, 64 hops max
 1 1000::1    2.173 ms   1.543 ms   2.214 ms
 2 1001::2    6.370 ms   4.353 ms   15.775 ms
 3 1002::a    1.892 ms   5.137 ms   9.492 ms

VPCS> █
```

Проверка с PC2: ping и trace до PC1

```
PC2-akabrikosov - PuTTY

VPCS> ping 1000::a

1000::a icmp6_seq=1 ttl=60 time=14.137 ms
1000::a icmp6_seq=2 ttl=60 time=17.220 ms
1000::a icmp6_seq=3 ttl=60 time=23.357 ms
1000::a icmp6_seq=4 ttl=60 time=17.335 ms
1000::a icmp6_seq=5 ttl=60 time=4.437 ms

VPCS> trace 1000::a

trace to 1000::a, 64 hops max
 1 1002::1    4.148 ms   2.395 ms   0.916 ms
 2 1001::1    3.206 ms   2.228 ms   13.861 ms
 3 1000::a    4.709 ms   3.316 ms   7.045 ms

VPCS> 
```

Трафик IPv6 при передаче через IPv4

No.	Time	Source	Destination	Protocol	Length	Info
25	10.113189	1002::a	1000::a	ICMPv6	194	Destination Unreachable (Port unreachable)[M
26	10.130695	1000::a	1002::a	UDP	146	65408 → 65409 Len=64
27	10.133913	1002::a	1000::a	ICMPv6	194	Destination Unreachable (Port unreachable)[M
28	33.238407	1002::a	1000::a	ICMPv6	138	Echo (ping) request id=0xcbf8, seq=1, hop li
29	33.241906	1000::a	1002::a	ICMPv6	138	Echo (ping) reply id=0xcbf8, seq=1, hop limi
30	34.260925	1002::a	1000::a	ICMPv6	138	Echo (ping) request id=0xcbf8, seq=2, hop li
31	34.272833	1000::a	1002::a	ICMPv6	138	Echo (ping) reply id=0xcbf8, seq=2, hop limi
32	35.281897	1002::a	1000::a	ICMPv6	138	Echo (ping) request id=0xcbf8, seq=3, hop li
33	35.290334	1000::a	1002::a	ICMPv6	138	Echo (ping) reply id=0xcbf8, seq=3, hop limi
34	36.310797	1002::a	1000::a	ICMPv6	138	Echo (ping) request id=0xcbf8, seq=4, hop li
35	36.322016	1000::a	1002::a	ICMPv6	138	Echo (ping) reply id=0xcbf8, seq=4, hop limi
36	37.345583	1002::a	1000::a	ICMPv6	138	Echo (ping) request id=0xcbf8, seq=5, hop li
37	37.347199	1000::a	1002::a	ICMPv6	138	Echo (ping) reply id=0xcbf8, seq=5, hop limi
38	37.394111	10.0.0.2	224.0.0.9	RIPv2	66	Response
39	38.357970	0c:24:7d:5a:00:00	0c:6d:97:11:00:01	ARP	60	Who has 10.0.0.1? Tell 10.0.0.2
40	38.367367	0c:6d:97:11:00:01	0c:24:7d:5a:00:00	ARP	60	10.0.0.1 is at 0c:6d:97:11:00:01

Рис. 25: Трафик при работе туннеля

Где видна информация о прохождении через туннель

The image shows a Wireshark packet capture analysis. The packet list on the left shows three packets. The selected packet is packet 31, an Internet Control Message Protocol (ICMP) Echo (ping) reply. The packet details pane on the right shows the following information:

- Frame 31: Packet, 138 bytes on wire (1104 bits), 138 bytes captured (1104 bits) on interface -, id 0
- Ethernet II, Src: 0c:6d:97:11:00:01 (0c:6d:97:11:00:01), Dst: 0c:24:7d:5a:00:00 (0c:24:7d:5a:00:00)
- Internet Protocol Version 4, Src: 10.0.0.1, Dst: 20.0.0.2
 - 0100 = Version: 4
 - 0101 = Header Length: 20 bytes (5)
 - Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
 - Total Length: 124
 - Identification: 0x9485 (38021)
 - 010. = Flags: 0x2, Don't fragment
 - ...0 0000 0000 0000 = Fragment Offset: 0
 - Time to Live: 64
 - Protocol: IPv6 (41)
 - Header Checksum: 0x87d1 [validation disabled]
 - [Header checksum status: Unverified]
 - Source Address: 10.0.0.1
 - Destination Address: 20.0.0.2
 - [Stream index: 0]
- Internet Protocol Version 6, Src: 1000::a, Dst: 1002::a
 - 0110 = Version: 6
 - 0000 0000 = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)
 - 0000 0000 0000 0000 = Flow Label: 0x000000
 - Payload Length: 64
 - Next Header: ICMPv6 (58)
 - Hop Limit: 61
 - Source Address: 1000::a
 - Destination Address: 1002::a
 - [Stream index: 0]
- Internet Control Message Protocol v6
 - Type: Echo (ping) reply (129)
 - Code: 0
 - Checksum: 0x9b61 [correct]

Выводы по работе

В GNS3 настроена сеть с VyOS и VPCS, выполнена адресация IPv4/IPv6 и проведена проверка связности. Динамическая маршрутизация IPv4 (RIP) обеспечила доступность удалённых IPv4-сетей. Туннель SIT и статические маршруты IPv6 обеспечили обмен трафиком между удалёнными IPv6-сегментами. Анализ Wireshark подтвердил обмен ARP/ICMP/RIP и наличие инкапсуляции IPv6-пакетов внутри IPv4 при передаче через туннель.